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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 11: Support Vector Machines and Clustering

LECTURE W11L1: Support Vector Machines – Part I

BY

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CONCEPTS COVERED

Support Vector Machines

- **Introduction**
- **Linearly separable case**
- **Example**

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Introduction to Support Vector Machines (SVM)

- Support vector machines (SVMs) have gained popularity in non-linear classification and regression problems since the mid-1990s.
- It is primarily recognized for its classification capabilities and is a kernel-based approach.
- Classification are made based on the decision boundaries that are developed using available data that may or may not be linearly separable.
- The method of SVM can also be used for non-linear regression problems that is known as Support Vector Regression (SVR).
- The SVM classifier for the two-class problem is developed in three stages:
  a) Maximum Margin Classifier: It was initially introduced to problems in which the two classes can be distinguished by a linear decision boundary, known as hyperplane.

Introduction to Support Vector Machines (SVM)

  b) Misclassification: The classifier is improved to consider misclassification in problems that are not separable by a linear decision boundary.
  c) Generalized Classifier: Through kernel method to include non-linear cases.

- A key advantage of SVMs is that their training does not suffer from the local-minimum problem commonly associated with MLP neural networks. Consequently, studies comparing a single SVM with a single MLP may report superior SVM performance.
- However, MLPs are usually trained as an ensemble to reduce this limitation. In regression problems, SVMs and ensemble MLPs with one or two hidden layers generally show comparable performance.
- With sufficiently large datasets, modern deep neural networks may outperform SVMs too.

Classification Frameworks of SVMs

There are four main classification frameworks of SVMs:

a) Linearly Separable Case: Applied when two distinct classes can be perfectly divided by a straight line or hyperplane.

b) Linearly Non-Separable Case: Utilized when perfect linear separation is impossible without errors.

c) Non-Linear Classification: Designed for complex datasets requiring curved or irregular decision boundaries.

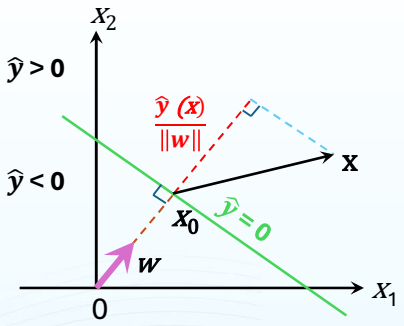
d) Multi-Class Classification: Extends the foundational binary (two-class) SVM architecture to accommodate three or more classes.

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Linearly Separable Case

- Let's assume that a dataset from the two classes can be separated by a hyperplane decision boundary.
- The equation that defines the separating hyperplane (as depicted in the figure) is as follows:

$$\hat{y}(x) = w^T x + w_0 = 0 \quad \ldots(1)$$



- The vector w is perpendicular to the hyperplane $\hat{y} = 0$. The hyperplane ($\hat{y} = 0$) is actually a straight line on this two-dimensional plane.
- The dashed line illustrates the component of the vector $x - x_0$ that is projected onto the w direction.

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Linearly Separable Case

- Any two points x_1 and x_2 , located on the hyperplane $\hat{y} = 0$ satisfy –

$$w^T(\mathrm{x}_1 - \mathrm{x}_2) = 0 \qquad \ldots(2)$$

- Hence the unit vector, normal to the $\hat{y} = 0$ hyperplane surface is given by :

$$\widehat{w} = \frac{\mathrm{w}}{\|\mathrm{w}\|} \qquad \ldots(3)$$

- The component of the vector x $-\mathrm{x}_0$ projected onto the $\widehat{\mathrm{w}}$ direction is given by –

$$\widehat{\boldsymbol{w}}^{\mathbf{T}}(\mathbf{x} - \mathbf{x_0}) = \frac{\boldsymbol{w}^{\mathbf{T}}(\mathrm{x}-\mathrm{x}_0)}{\|\boldsymbol{w}\|} = \frac{\boldsymbol{w}^{\mathbf{T}}\mathbf{x}+w_0}{\|\boldsymbol{w}\|} = \frac{\hat{y}(\mathbf{x})}{\|\boldsymbol{w}\|} \qquad \ldots(4)$$

- Thus, the sign of $\hat{y}(\mathbf{x})$ is indicative of the side of the hyperplane on which **x** is located, and it is proportional to the normal distance between **x** and the hyperplane $\hat{y}(\mathbf{x}) = 0$.

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Linearly Separable Case

- It is assumed that the training dataset is composed of predictors x_n and target data y_n $(n = 1, \ldots, N)$. The value of y_n is either −1 or +1, as there are two classes.
- When a new predictor vector *x* becomes available, it is classified according to the sign of the function,

$$\hat{y}(\mathrm{x}) = \mathrm{w}^{\mathrm{T}}\mathrm{x} + \mathrm{w}_0 \qquad \ldots(5)$$

- The normal distance from a point x_n to the decision boundary $\hat{y} = 0$ is, according to (4), given by –

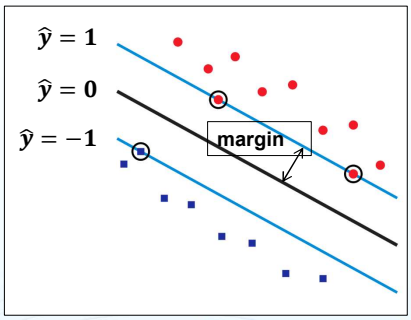
$$\frac{y_n\hat{y}(\mathbf{x}_n)}{\|\mathbf{w}\|} = \frac{y_n(\mathbf{w}^T\mathbf{x}_n+w_0)}{\|\mathbf{w}\|} \qquad \ldots(6)$$

where the correct sign (+1 or −1) is contributed by y_n to make sure that the distance given by (6) is non-negative.

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Linearly Separable Case

- The dataset depicted in the accompanying figure is separable by a hyperplane decision boundary $\hat{y} = 0$ and consists of two classes (represented by solid circles and squares).
- The margin is determined by the distance of the nearest point(s) x_n in the dataset and is maximized.
- Circled points are support vectors, that is, data points used in determining the margins $\hat{y} = \pm 1$.
- The decision boundary is established by a maximum margin classifier, which maximizes the margin l by identifying the optimal values of w and w_0.



Linearly Separable Case

- Therefore, the optimization problem is as follows:

$$\max_{\boldsymbol{w},w_0} l \text{ subject to } \frac{y_n(\boldsymbol{w}^T \mathrm{x}_n + w_0)}{\|\boldsymbol{w}\|} \geq l, \ (n = 1, \ldots, N) \qquad \ldots(7)$$

where the constraint guarantees that no data points are contained within the margins. If we choose $\|w\| = l^{-1}$, then the constraint in (7) becomes:

$$y_n(\boldsymbol{w}^T \mathrm{x}_n + w_0) \geq 1, \quad (n = 1, \ldots, N) \qquad \ldots(8)$$

- Maximizing the margin l is equivalent to minimizing w, which is equivalent to solving:

$$\min_{\boldsymbol{w},w_0} \frac{1}{2} \|\boldsymbol{w}\|^2 \qquad \ldots(9)$$

subject to the constraint (8). This optimization of a quadratic function subject to constraints is referred to as a quadratic programming problem.

Linearly Separable Case

- This type of optimization problem can be solved with an inequality constraint through a generalization of the Lagrange multiplier method.
- A Lagrange multiplier $\lambda_n \geq 0$ is introduced for each of the N constraints in (8), resulting in the Lagrange function L taking the form

$$L(w, w_0, \lambda) = \frac{1}{2}\|w\|^2 - \sum_{n=1}^{N} \lambda_n [y_n(w^T x_n + w_0) - 1] \quad \ldots(10)$$

with $\lambda = (\lambda_1, \lambda_2, \ldots \lambda_n)^T$.

- The derivatives of L with respect to w and w_0 are set to zero resulting in the following :

$$w = \sum_{n=1}^{N} \lambda_n y_n x_n \quad \ldots(11)$$

$$0 = \sum_{n=1}^{N} \lambda_n y_n \quad \ldots(12)$$

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Linearly Separable Case

- By substituting these into (10), we can express L exclusively in terms of the Lagrange multipliers, that is,

$$L_D(\lambda) = \sum_{n=1}^{N} \lambda_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} \lambda_n \lambda_j y_n y_j x_n^T x_j \quad \ldots(13)$$

subject to $\lambda_n \geq 0$ and (12), with L_D referred to as the dual Lagrangian. Consequently, The dual problem (13) is also a quadratic programming problem with N variables from λ.

- The primary rationale for employing the dual problem instead of the primal problem is that, in the subsequent stage, to generalize the linear classifier to a non-linear classifier, SVM will invoke kernels, that is, **x** will be replaced by $\varphi(\mathbf{x})$ in a feature space of dimension M; thus, m will be replaced by M, which is usually much larger or even infinite.

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Linearly Separable Case

- Our constrained optimization problem satisfies the KKT conditions,

$$\lambda_n \geq 0, \quad \ldots(14)$$

$$y_n(w^T x_n + w_0) - 1 \geq 0, \quad \ldots(15)$$

$$\lambda_n[y_n(w^T x_n + w_0) - 1] = 0 \quad \ldots(16)$$

- Combining (5) and (11), we have the following formula for classifying a new data point *x*:

$$\hat{y}(x) = \sum_{n=1}^{N} \lambda_n y_n x_n^T x + w_0 \quad \ldots(17)$$

where the class (+1 or −1) is decided by the sign of $\hat{y}(x)$.

- From (5), the KKT condition (16) can be rewritten as –

$$\lambda_n[y_n \hat{y}(x_n) - 1] = 0 \quad \ldots(18)$$

- Data points with $\lambda_n > 0$ are the support vectors and they satisfy $y_n \hat{y}(x_n) = 1$, that is, they lie exactly on the maximum margins.

Linearly Separable Case

- The support vectors were located using the entire training dataset, despite the fact that only support vectors determine the margins and the decision boundary.
- Nevertheless, the support vectors are the sole requirement for the classification of new data by (17) once the model has been trained, and the remaining data (usually the majority of data points) can be disregarded.
- Sparsity or Sparseness is the desirable property of being able to disregard the majority of data points, which significantly reduces computational costs.
- To obtain the value for w_0, we use the fact that for any support vector x_j, $y_j \hat{y}(x_j) = 1$, that is,

$$y_j\left(\sum_{n=1}^{N} \lambda_n y_n \mathbf{x}_n^T \mathbf{x}_j + w_0\right) = 1 \quad \ldots(19)$$

Linearly Separable Case

- As $y_j = \pm 1$, $y_j^{-1} = y_j$, so

$$w_0 = y_j - \sum_{n=1}^{N} \lambda_n y_n \mathbf{x}_n^T \mathbf{x}_j \qquad (20)$$

(where the summation need only be over n for which λ_n is non-zero).

- Since each support vector $\mathbf{x}_j$ gives an estimate of w_0, all the estimates are usually averaged together to provide the most accurate estimate of w_0.

Example:

A researcher is using a Support Vector Machine (SVM) model to classify a dataset into two distinct categories: Class -1 and Class +1. The classification relies on two features, x_1 and x_2.

The training dataset contains the following 4 points:

P_1: $x_1 = (1, 1)^T$, Class $y_1 = -1$; P_2: $x_2 = (1, 2)^T$, Class $y_2 = -1$

P_3: $x_3 = (3, 3)^T$, Class $y_3 = +1$; P_4: $x_4 = (4, 3)^T$, Class $y_4 = +1$

Identify the support vectors, determine the optimal separating hyperplane (decision boundary), and calculate the margin.

A new data point is recorded with parameters $x_{new} = (2.5, 2)^T$. Using the trained SVM model, classify this new point.

Solution:

Step 1: Identifying the Support Vectors and Lagrange Multipliers Constraint

By evaluating the spatial coordinates of the dataset, P_2 (Class - 1) and P_3 (Class + 1) represent the closest boundary points between the two classes. P_1 and P_4 are located further away from the margin.

According to the Karush-Kuhn-Tucker (KKT) conditions, points lying strictly outside the margin will have their Lagrange multipliers (λ) driven to exactly zero.

Therefore:

$$\lambda_1 = 0$$
$$\lambda_4 = 0$$

The support vectors are P_2 and P_3, meaning $\lambda_2 > 0$ and $\lambda_3 > 0$.

Solution:

According to Equation (12) from the KKT conditions:

$$0 = \sum_{n=1}^{N} \lambda_n y_n \Rightarrow \lambda_2(-1) + \lambda_3(1) = 0 \Rightarrow \lambda_2 = \lambda_3$$

Let us denote both as λ.

Step 2: Expressing the Weight Vector (w) algebraically

Equation (11) defines the weight vector in terms of the support vectors:

$$w = \sum_{n=1}^{N} \lambda_n y_n x_n = \lambda(-1)\begin{pmatrix}1\\2\end{pmatrix} + \lambda(1)\begin{pmatrix}3\\3\end{pmatrix} = \begin{pmatrix}-\lambda\\-2\lambda\end{pmatrix}\begin{pmatrix}3\lambda\\3\lambda\end{pmatrix} = \begin{pmatrix}2\lambda\\\lambda\end{pmatrix}$$

Solution:

Thus, $w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 2\lambda \\ \lambda \end{pmatrix} \Rightarrow w_1 = 2\lambda$ and $w_2 = \lambda$

Step 3: Applying the KKT Margin Condition to create a system of equations

Equation (16) provides the KKT condition for the margins:

$$\lambda_n[y_n(w^T x_n + w_0) - 1] = 0$$

Because $\lambda > 0$ for our support vectors, the term inside the brackets must equal zero. Therefore, $y_n(w^T x_n + w_0) = 1$.

We can plug our two support vectors into this equation:

For P_2 where $y_2 = -1$:

$$(-1)(w_1(1) + w_2(2) + w_0) = 1$$
$$\Rightarrow w_1 + 2w_2 + w_0 = -1$$



Solution:

For P_3 where $y_3 = +1$:

$$(1)(w_1(3) + w_2(3) + w_0) = 1$$
$$\Rightarrow 3w_1 + 3w_2 + w_0 = 1$$

Step 4: Solving the algebraic system

Now, substituting the weight vector components $w_1 = 2\lambda$ and $w_2 = \lambda$ from Step 2 into the two margin equations:

For P_2 : $w_1 + 2w_2 + w_0 = -1 \Rightarrow 2\lambda + 2(\lambda) + w_0 = -1$

$$\Rightarrow 4\lambda + w_0 = -1$$

For P_3: $3w_1 + 3w_2 + w_0 = 1 \Rightarrow 3(2\lambda) + 3(\lambda) + w_0 = 1$

$$\Rightarrow 9\lambda + w_0 = 1$$



Solution:

Subtracting the first equation from the second to eliminate w_0 :

$$9\lambda + w_0 - (4\lambda + w_0) = 1 - (-1) \Rightarrow 5\lambda = 2 \Rightarrow \lambda = \frac{2}{5} = 0.4$$

Step 5: Calculating w and w_0

Now that we have λ, we plug it back in to find the hyperplane parameters:

$w_1 = 2\lambda = 2(0.4) = 0.8$ and $w_2 = \lambda = 0.4$

So, $w = \begin{pmatrix} 0.8 \\ 0.4 \end{pmatrix}$

Also, as established previously,

$$4\lambda + w_0 = -1 \Rightarrow 4(0.4) + w_0 = -1$$

$$\Rightarrow w_0 = -1 - 4(0.4) = -1 - 1.6 = -2.6 \Rightarrow w_0 = -2.6$$

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Solution:

Therefore, the hyperplane equation is:

$$\hat{y}(x) = w^T x + w_0 = 0 \Rightarrow 0.8x_1 + 0.4x_2 - 2.6 = 0$$

Margin: $l = \frac{1}{\|w\|} = \frac{1}{\sqrt{0.8^2 + 0.4^2}} = \frac{1}{\sqrt{0.8}} \approx 1.12$

Step 6: Classifying the new point

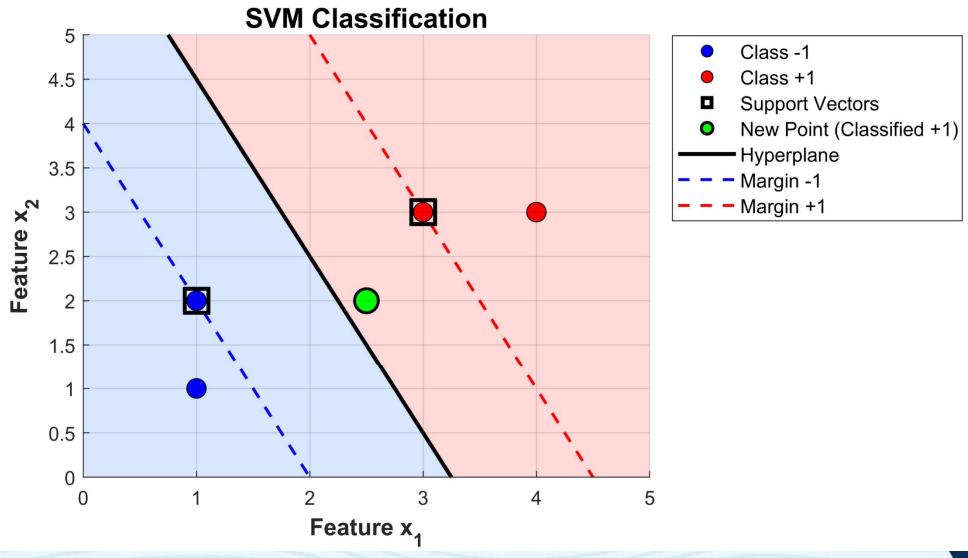
The new point is x_{new} = $(2.5, 2)^T$. We classify it according to the sign of the function $\hat{y}(x) = w^T x + w_0$ (Equation 5)

$$\hat{y}(x_{new}) = 0.8(2.5) + 0.4(2) - 2.6 = 2 + 0.8 - 2.6 = 0.2$$

Because the result $0.2 > 0$, the SVM classifies the new point as +1.

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Solution:


SVM Classification
Feature x2
Feature x1
Class -1
Class +1
Support Vectors
New Point (Classified +1)
Hyperplane
Margin -1
Margin +1


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LECTURE W11L2: Support Vector Machines – Part II

BY

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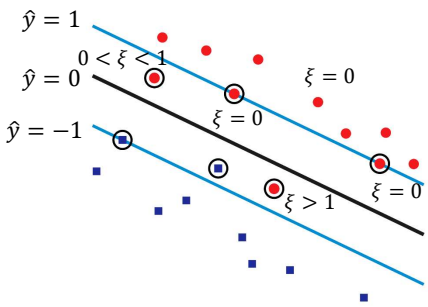
CONCEPTS COVERED

Support Vector Machines

- **Linearly non-separable case**
- **Non-Linear Case**
- **Example**

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Linearly Non-Separable Case



- This is the situation in which the two classes are not separable by a hyperplane. In other words, regardless of the effort exerted, one or more data points will be misclassified and end up on the incorrect side of the hyperplane.
- Consequently, the classifier must be generalized to accommodate misclassification.
- We introduce slack variables, $\xi_n \geq 0$, $n = 1, \ldots, N$. Each training data point x_n is assigned a slack variable ξ_n.
- The dataset depicted in the accompanying figure is not separable by a hyperplane boundary. Slack variables $\xi_n \geq 0$ are introduced, with $\xi_n = 0$ for data points lying on or within the correct margin, $\xi_n > 1$ for points lying to the wrong side of the decision boundary. Support vectors are circled.

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Linearly Non-Separable Case

- The slack variables are defined by $\xi_n = 0$ for all x_n that are lying on or within the correct margin boundary, and $\xi_n = |\, y_n - \hat{y}(x_n)|$ for all other points.
- Therefore, the slack variable quantifies the distance a data point extends beyond the correct margin (as illustrated in prev. figure).
- A data point that lies directly on the decision boundary $\hat{y}(x_n) = 1$ will have $\xi_n = 1$, whereas, $\xi_n > 1$ corresponds to points that are misclassified, i.e., located on the wrong side of the decision boundary.
- Points with $0 < \xi_n \leq 1$ extend beyond the correct margin but not enough to cross the decision boundary to be misclassified.
- The constraint (used in linearly separable case) is revised to accommodate data points that extend beyond the correct margin, i.e.,

$$y_n(w^T x_n + w_0) \geq 1 - \xi_n, \; (n = 1, \ldots, N) \qquad (1)$$

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Linearly Non-Separable Case

- The minimization problem is –

$$\min_{w,w_0}\left(\frac{1}{2}\|w\|^2 + C\sum_{n=1}^{N}\xi_n\right) \qquad (2)$$

- The optimization aims to maximize the margin while allowing some slack for misclassification. C is a hyperparameter, with a smaller value allowing more slack.
- If we divide the expression in (2) by C, the second term can be viewed as an error term, while the first term can be viewed as a weight penalty term, with $C-1$ as the weight penalty parameter – analogous to the regularization of NN models where we have a mean squared error (MSE) term plus a weight penalty term in the objective function.
- Contrary to the MSE term, which is quadratic, the objective function's response to misclassification is only linearly related to ξ_n in (2).



Linearly Non-Separable Case

- Given that any misclassified point has $\xi_n > 1$, $\sum_n \xi_n$ can be seen as providing an upper limit on the number of misclassified points.
- To optimize (2) subject to constraint (1) and $\xi_n \geq 0$, the method of Lagrange multipliers is employed, with the Lagrange function now being:

$$L = \frac{1}{2}\|w\|^2 + C\sum_{n=1}^{N}\xi_n - \sum_{n=1}^{N}\lambda_n[y_n(w^T x_n + w_0) - 1 + \xi_n] - \sum_{n=1}^{N}\mu_n\xi_n \qquad (3)$$

with $\lambda_n \geq 0$ and $\mu_n \geq 0$ $(n = 1, \ldots, N)$ the Lagrange multipliers.

- Setting the derivatives of L with respect to w, w_0 and ξ_n to zero, results in, respectively,

$$w = \sum_{n=1}^{N}\lambda_n y_n x_n\,, \qquad (4)$$

$$0 = \sum_{n=1}^{N}\lambda_n y_n\,, \qquad (5)$$

$$\lambda_n = C - \mu_n \qquad (6)$$



Linearly Non-Separable Case

- Substituting these into (3) again allows L to be expressed exclusively in terms of the Lagrange multipliers λ, that is,

$$L_D(\lambda) = \sum_{n=1}^{N} \lambda_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} \lambda_n \lambda_j y_n y_j x_n^T x_j \qquad (7)$$

  with L_D the dual Lagrangian. As $\lambda_n \geq 0$ and $\mu_n \geq 0$, (6) implies

$$0 \leq \lambda_n \leq C \qquad (8)$$

- Additionally, the constraints from the KKT conditions for $n = 1, \ldots, N$ are:

$$y_n(w^T x_n + w_0) - 1 + \xi_n \geq 0, \qquad (9)$$

$$\lambda_n[y_n(w^T x_n + w_0) - 1 + \xi_n] = 0, \qquad (10)$$

$$\mu_n \xi_n = 0 \qquad (11)$$

- In the calculation for w in (4), the data points with $\lambda_n \neq 0$ contributed in the summation, so these points (with $\lambda_n > 0$) are the support vectors.

Linearly Non-Separable Case

- The value for w_0 can be found from (10) using any of the support vectors located on the margin ($\xi_n = 0$),and typically the values obtained from the individual vectors are averaged together to give the best estimate.
- Optimizing the dual problem (7) is a simpler quadratic programming problem than the primal (3) and can be easily solved with standard methods.
- The hyperparameter C is not solved by the optimization. Typically, a portion of validation data is allocated to evaluate the efficacy of many models trained with varying values of C in order to ascertain the best value for C.
- The classification with the new data x is based on the sign of $\hat{y}(x)$.

Non-Linear Case

- Non-linear classification is done by carrying out linear classification using $\varphi(\mathrm{x})$ data in a feature space rather than the input *x* data where φ represents the non-linear function that maps the original input space to the feature space that is,

$$\hat{\mathrm{y}}(\mathrm{x}) = w^T \phi(\mathrm{x}) + w_0 \qquad (12)$$

- The dual Lagrangian L_D for this revised problem is derived from (7) merely substituting x by $\varphi(\mathrm{x})$, result in :

$$L_D(\lambda) = \sum_{n=1}^{N} \lambda_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} \lambda_n \lambda_j \mathrm{y}_n \mathrm{y}_j \phi^T(\mathrm{x}_n) \phi(\mathrm{x}_j) \qquad (13)$$

- Classification is based on the sign of $\hat{\mathrm{y}}(x)$, with $\hat{\mathrm{y}}(\mathrm{x})$ being:

$$\hat{\mathrm{y}}(x) = \sum_{n=1}^{N} \lambda_n \mathrm{y}_n \ \phi^T(\mathrm{x}_n) \phi(\mathrm{x}) + w_0 \qquad (14)$$



Non-Linear Case

- Since the dimension of the feature space may be very high or even infinite, calculations involving the inner product $\varphi^T(\mathrm{x})\varphi(\mathrm{x}')$ are feasible only due to the kernel trick, where kernel function K is:

$$K(\mathrm{x}, \mathrm{x}') = \phi^T(\mathrm{x}) \phi(\mathrm{x}') \qquad (15)$$

- K is used to eliminate the need of direct calculation of the inner product, with the Gaussian or radial basis function (RBF) kernel being the most prevalent,

$$K(\mathrm{x}, \mathrm{x}') = \exp\left(- \frac{\|\mathrm{x} - \mathrm{x}'\|^2}{2\sigma^2} \right) \qquad (16)$$

- Under the kernel approach, (14) becomes

$$\mathrm{y}(\mathrm{x}) = \sum_{n=1}^{N} \lambda_n \mathrm{y}_n \ K(\mathrm{x}_n, \mathrm{x}) + w_0 \qquad (17)$$

- and (13) becomes:

$$L_D(\lambda) = \sum_{n=1}^{N} \lambda_n - \frac{1}{2} \sum_{n=1}^{N} \sum_{j=1}^{N} \lambda_n \lambda_j \mathrm{y}_n \mathrm{y}_j \ K(\mathrm{x}_n, \mathrm{x}_j) \qquad (18)$$



Non- Linear Case

- Optimizing the dual problem (18) is yet again a quadratic programming problem.
- Given that the objective function is only quadratic, and with the constraints being linear, there is no local minima problem, that is, unlike MLP NN models, the minimum found by SVM is the global minimum.
- The hyperparameters C and σ (assuming the RBF kernel (16) is used) are not derived from the optimization process.
- Various models with different values of C and σ are trained, and from their classification performance over validation data, optimal values for C and σ are determined.
- Another alternative is to employ Bayesian optimization to determine the hyperparameters.

Example:

Suppose you are analyzing a linearly non-separable dataset using a Support Vector Machine (SVM). The optimization process has yielded three support vectors (where $\lambda_n > 0$) that dictate the decision boundary. The details of these support vectors are:

SV1: $x_1 = (2, 0)^T$, Class $y_1 = +1$, $\lambda_1 = 0.5$; SV2: $x_2 = (0, 2)^T$, Class $y_2 = +1$, $\lambda_2 = 0.5$

SV3: $x_3 = (0, 0)^T$, Class $y_3 = -1$, $\lambda_3 = 1$;

Assume all given support vectors lie exactly on the margin ($\xi = 0$).

a) Calculate the optimal weight vector w and the bias w_0.

b) Given four new training data points, calculate the required slack variable ξ_n for each and determine its classification status:

Example:

Point A: $x_A = (2, 2)^T$, Class $y_A = +1$

Point B : $x_B = (1, 0)^T$, Class $y_B = +1$

Point C : $x_C = (1, 0.5)^T$, Class $y_C = +1$

Point D : $x_D = (2, 1)^T$, Class $y_D = -1$

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Solution:

Step 1: Calculating the Weight Vector (*w*)

The weight vector is calculated using the formula derived from setting the derivative of the Lagrangian function to zero (equation (4)):

$$w = \sum_{n=1}^{N} \lambda_n y_n x_n = \lambda_1 y_1 x_1 + \lambda_2 y_2 x_2 + \lambda_3 y_3 x_3$$

$$\Rightarrow w = (0.5)(1)\begin{pmatrix}2\\0\end{pmatrix} + (0.5)(1)\begin{pmatrix}0\\2\end{pmatrix} + (1.0)(-1)\begin{pmatrix}0\\0\end{pmatrix}$$

$$\Rightarrow w = \begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix} - \begin{pmatrix}0\\0\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}$$

$$\Rightarrow w = \begin{pmatrix}1\\1\end{pmatrix}$$

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Solution:

Step 2: Calculating the Bias (w_0)

The bias is obtained using any of the support vectors lying on the margin ($\xi_n = 0$) using the formula :

$$w_0 = y_j - \sum_{n=1}^{N} \lambda_n y_n x_n^T x_j = y_j - w^T x_j$$

Usually, the values obtained from the individual vectors are averaged together to give the best estimate. Let's verify this across all three support vectors:

Using SV1: $w_0 = 1 - \left((1,1).(2,0)^T\right) = 1 - (2+0) = -1$

Using SV2: $w_0 = 1 - \left((1,1).(0,2)^T\right) = 1 - (0+2) = -1$

Using SV3: $w_0 = (-1) - \left((1,1).(0,0)^T\right) = (-1) - (0+0) = -1$

The average is clearly -1. Thus, the decision boundary function is: $w^T x_j + w_0 = 0 \Rightarrow x_1 + x_2 - 1 = 0$

Solution:

Step 3: Calculating Slack Variables (ξ_n)

Recalling equation (1):

$$y_n(w^T x_n + w_0) \geq 1 - \xi_n, \ (n = 1, \ldots, N)$$

As we know that, a point is considered to be on or within the correct margin boundary if $y_n \hat{y}(x_n) \geq 1$. If this is true, $\xi_n = 0$. If it is not true, we use the formula :

$\xi_n = |\, y_n - \hat{y}(x_n)|$

Let's evaluate the four test points based on this exact rule:

Point A (x_A = (2, 2)^T, Class y_A = +1):

Boundary value: $\hat{y}(x_A) = 2 + 2 - 1 = 3$

Margin check: $y_A \hat{y}(x_A) = 1 \times 3 = 3$ (This is ≥ 1)

So, the point lies within the correct margin boundary, i.e., $\xi_A = 0$

Conclusion: Point A is correctly classified.

Solution:

Point B (x_B = (1, 0)^T, Class y_B = +1):

Boundary value: $\hat{y}(x_B) = 1 + 0 - 1 = 0$

Margin check: $y_B\hat{y}(x_B) = 1 \times 0 = 0$ (This is not ≥ 1)

Using the formula for other points, $\xi_B = |y_B - \hat{y}(x_B)| = |1 - 0| = 1$

Conclusion: A data point with $\hat{y}(x_n) = 0$ lies right on the decision boundary.

Point C (x_C = (1, 0.5)^T, Class y_C = +1):

Boundary value: $\hat{y}(x_C) = 1 + 0.5 - 1 = 0.5$

Margin check: $y_C\hat{y}(x_C) = 1 \times 0.5 = 0.5$ (This is not ≥ 1)

Using the formula for other points, $\xi_C = |y_C - \hat{y}(x_C)| = |1 - 0.5| = 0.5$

Conclusion: Because, $0 < \xi_C < 1$, the point protrudes beyond the margin but not enough to be misclassified.

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Solution:

Point D (x_D = (2, 1)^T, Class y_D = -1):

Boundary value: $\hat{y}(x_D) = 2 + 1 - 1 = 2$

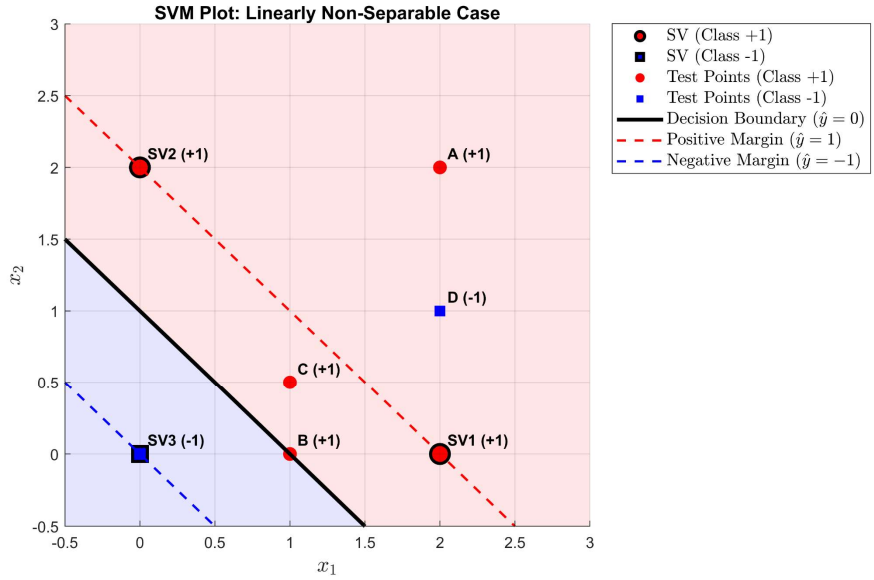
Margin check: $y_B\hat{y}(x_B) = (-1) \times 2 = -2$ (This is not ≥ 1)

Using the formula for other points, $\xi_D = |y_D - \hat{y}(x_D)| = |-1 - 2| = |-3| = 3$

Conclusion: Because, $\xi_D > 1$, it corresponds to a point lying to the wrong side of the decision boundary and is misclassified.

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Solution:


SVM Plot: Linearly Non-Separable Case
SV (Class +1)
SV (Class -1)
Test Points (Class +1)
Test Points (Class -1)
Decision Boundary ($\hat{y} = 0$)
Positive Margin ($\hat{y} = 1$)
Negative Margin ($\hat{y} = -1$)
SV2 (+1)
A (+1)
D (-1)
C (+1)
SV3 (-1)
B (+1)
SV1 (+1)
x_1
x_2


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 11: Support Vector Machines and Clustering

LECTURE W11L3: Support Vector Machines – Part III

BY

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CONCEPTS COVERED

Support Vector Machines

- **Multi-class classification case**
- **Examples**

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Multi-class Classification

- Support vector machine was initially developed for two-class (i.e., binary) problems.
- The simplest extension to multi-class problems is via the one-versus-the rest approach: that is, given c classes, we train c SVM binary classifiers $\hat{y}^{(k)}(\mathrm{x})$ $(k = 1, \ldots, c)$ where the target data for class k are set to +1 and for all remaining classes to −1.
- New input data ***x*** are then classified into class j if $\hat{y}^{(j)}(\mathrm{x})$ reaches the highest value among $\hat{y}^{(k)}(\mathrm{x})$ $(k = 1, \ldots, c)$.
- There are two criticisms of this heuristic approach:
  i. Since the c binary classifiers were trained on different problems, there is no assurance that the outputs $\hat{y}^{(k)}$ are on the same scale for fair comparison between the c binary classifiers.

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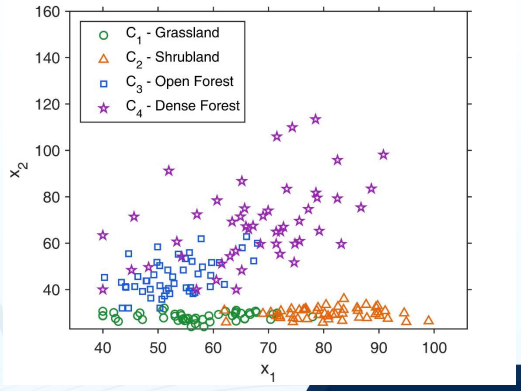
Multi-class Classification

ii. if numerous classes exist, then, the +1 target data may be significantly outnumbered by the −1 target data.

- A more effective approach is pairwise classification or one-versus-one.
  - A binary classifier is trained for each possible pair of classes, resulting in total of $c(c-1)/2$ binary classifiers, with each classifier using training data from only class k and class l $(k \neq l)$.
  - With new data ***x***, all the binary classifiers are applied, and if class j receives the highest number of 'yes votes' from the classifiers, then ***x*** is classified as belonging to class j.

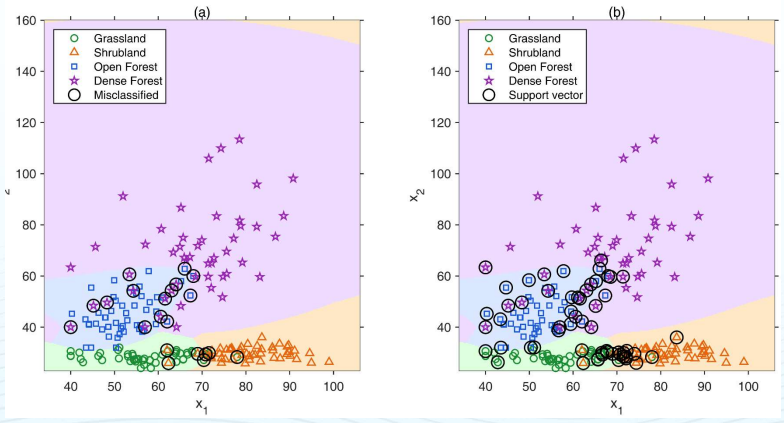
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Multi-class Classification



- For example, we consider a training dataset of forest cover that includes two predictor variables x_1 and x_2 derived from two spectral bands and four classes $(C_1, \ldots, C_4)$ of land surface (given figure).
- We apply the one-versus-one SVM classifier with Gaussian kernel on this dataset.
- Out of 200 training data points, 50 are support vectors (Fig. in the next slide), which are used to determine the boundaries of the decision regions.
- The misclassification rate is 10.5% for the training data and 21% for the test data.

Multi-class Classification



One-versus-one classification by SVM

- Decision regions for the four classes are shown by different background shading.
- Circled data points in (a) are the misclassified data and in (b) the support vectors.

Example – 1:

A satellite image is classified into three land-cover classes $(c = 3)$:

$C_1 =$ Water; $C_2 =$ Vegetation; and $C_3 =$ Built-up.

Two normalized spectral variables x_1 and x_2 are used.

Using the One-Versus-One approach, binary classifiers were trained for every possible pair of classes. The total number of classifiers is $c(c-1)/2 = 3$. The decision boundaries for these classifiers are given below:

Classifier Pair	Decision function $f_{ij}(x)$	Rule (vote)
C_1 vs C_2	$f_{12}(\mathrm{x}) = 2\mathrm{x}_1 + \mathrm{x}_2 - 6$	If $f > 0$, vote C_1; else vote C_2
C_1 vs C_3	$f_{13}(\mathrm{x}) = \mathrm{x}_1 + 2\mathrm{x}_2 - 6$	If $f > 0$, vote C_1; else vote C_3
C_2 vs C_3	$f_{23}(\mathrm{x}) = \mathrm{x}_2 - \mathrm{x}_1$	If $f > 0$, vote C_2; else vote C_3

Example – 1:

Given a new unclassified pixel with spectral features: $x = (1,2)$

a) Calculate the output for each of the three binary classifiers.

b) Determine which class receives the vote from each classifier.

c) Conclude the final classification for the pixel based on the majority vote.

Solution:

Given that the new unclassified pixel has spectral features: $x=(1,2)$, i.e., the feature values are: $x_1 = 1$ and $x_2 = 2$.

a) Evaluating each binary classifier:

Substituting the feature values $x_1 = 1$ and $x_2 = 2$ into the equations for each trained classifier:

Evaluating C_1 vs C_2:

$$f_{12}(x) = 2x_1 + x_2 - 6 = 2(1) + 2 - 6 = 4 - 6 = -2$$

Evaluating C_1 vs C_3:

$$f_{13}(x) = x_1 + 2x_2 - 6 = 1 + 2(2) - 6 = 5 - 6 = -1$$

Evaluating C_2 vs C_3:

$$f_{23}(x) = x_2 - x_1 = 2 - 1 = 1$$

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Solution:

b) Determinig the votes from each classifier:

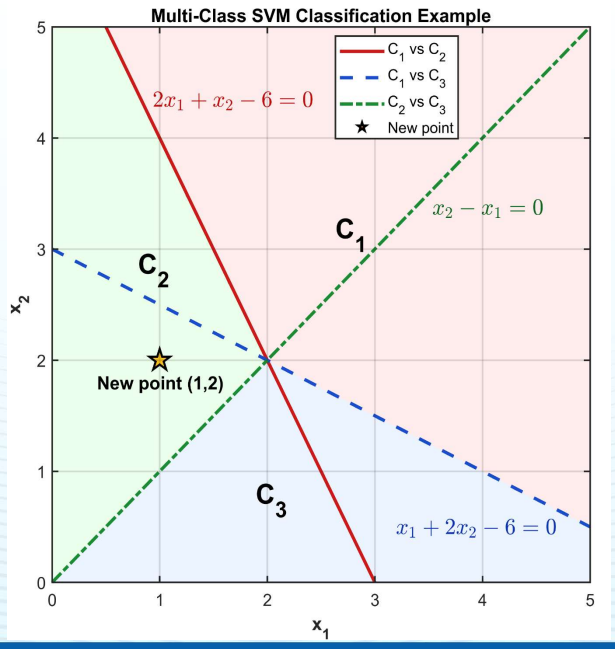
Classifier Pair	Decision function value	Rule (vote)	Vote Cast
C_1 vs C_2	$f_{12}(x) = -2$	If > 0, vote C_1; else vote C_2	$\boldsymbol{C_2}$
C_1 vs C_3	$f_{13}(x) = -1$	If > 0, vote C_1; else vote C_3	$\boldsymbol{C_3}$
C_2 vs C_3	$f_{23}(x) = 1$	If > 0, vote C_2; else vote C_3	$\boldsymbol{C_2}$

c) Classification of new pixel:

As per the votes casted, we can see that, the received votes are: $\boldsymbol{C_1 = 0, C_2 = 2\ and\ C_3 = 1}$.

As the highest no. of votes were received by $\boldsymbol{C_2}$, so, the One-vs-One SVM classifies the point as Class $\boldsymbol{C_2}$ (Vegetation).

Solution:


Multi-Class SVM Classification Example
C1 vs C2
C1 vs C3
C2 vs C3
New point
$2x_1 + x_2 - 6 = 0$
$x_2 - x_1 = 0$
$x_1 + 2x_2 - 6 = 0$
C1
C2
C3
New point (1,2)
x1
x2


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Statistical and Machine Learning Methods: Basics and Applications

MODULE 11: Support Vector Machines and Clustering

LECTURE W11L4: Clustering – Part I

BY

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CONCEPTS COVERED

Clustering

- **Distance Measure**
- **Model Evaluation**

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Clustering

- Clustering is an unsupervised learning approach utilized to discover hidden patterns within unlabeled datasets.
- This technique, groups objects (i.e. data points) into clusters, ensuring that objects within a cluster share greater similarity with one another than with objects in different groups.
- Clustering methods, are primarily categorized into hierarchical and non-hierarchical methods.
- In non-hierarchical clustering, such as K-means clustering, the relationship between clusters is not determined.
- In contrast, hierarchical clustering repeatedly links the pair of closest clusters until all data points are included in the hierarchy, which structurally resembles an inverted tree.

Clustering

- Although hierarchical clustering provides deeper insights into cluster relationships, it is highly susceptible to noise and generally falls short of the performance achieved by non-hierarchical clustering methods.
- Non-hierarchical clustering methods are broadly classified into two categories, hard clustering and soft clustering methods.
- In hard clustering, an object either belongs to a cluster or it does not, that is, its membership in a cluster is indicated by a binary number, 1 or 0.
- Under soft clustering (fuzzy clustering), an object can belong to multiple clusters with a different probability for each cluster.
- Clustering methods are widely used across environmental sciences to analyze climate zones, hydrology, and ecology. However, a significant limitation is that different clustering techniques may yield varying numbers of clusters.

Distance Measure

- A distance (also called dissimilarity) measure is required to evaluate how close two objects are, and there are many choices.
- The most widely used measure is the Euclidean distance, where for two data points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ (both being m-dimensional vectors), the distance is:

$$d_{ij} = \left[\sum_{l=1}^{m}\left(x_{il} - x_{jl}\right)^2\right]^{1/2} \quad (1)$$

with x_{il} and x_{jl} $(l = 1, \ldots, m)$ denoting the elements of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

- Because calculating square roots is computationally expensive, the squared Euclidean distance is frequently used as a substitute metric, defined as:

$$d_{ij} = \sum_{l=1}^{m}\left(x_{il} - x_{jl}\right)^2 \quad (2)$$

Distance Measure

- For K-means clustering, there is no difference between using Euclidean distance and squared Euclidean distance, so the squared variant is preferred due to its higher computational efficiency.
- For other clustering methods, the two distance measures may yield different outcomes.
- Furthermore, if $\boldsymbol{x}$ contains variables with different units (e.g. temperature and specific humidity), standardizing the data is necessary before performing clustering, otherwise the distance calculations will be influenced by the variable(s) with large standard deviation.
- More generally, the Minkowski distance,

$$d_{ij} = \left[\sum_{l=1}^{m}\left|x_{il} - x_{jl}\right|^{\lambda}\right]^{1/\lambda} \quad (3)$$

which reduces to the Euclidean distance when $\lambda = 2$.

Distance Measure

- When $\lambda = 1$, the Minkowski distance simplifies to the city-block distance (or Manhattan distance),

$$d_{ij} = \sum_{l=1}^{m} |x_{il} - x_{jl}| \qquad (4)$$

- Given N data points, d_{ij} form the elements of an $N \times N$ distance or dissimilarity matrix D.
- For ordinal variables, for example low, medium, high, it is common to code the values as real numbers, such as 1, 2, 3, to enable the application of the aforementioned distance measures.

Distance Measure

- For categorial variables, such as red, green, blue; the Hamming distance is most commonly used,

$$d_{ij} = \sum_{l=1}^{m} I(x_{il} \neq x_{jl}), \qquad (5)$$

where,

$$I(x_{il} \neq x_{jl}) = \begin{cases} 1 & \text{if } x_{il} \neq x_{jl} \\ 0 & \text{if } x_{il} = x_{jl} \end{cases} \qquad (6)$$

that is, one simply counts the number of non-matching elements between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

Model Evaluation

- The biggest question in clustering is how to find the optimal number of clusters, K.
- Fundamentally, the issue lies in identifying the most effective configuration after evaluating a model across a range of K values.
- Unlike supervised learning, the lack of target data in unsupervised learning makes it difficult to evaluate model performance.
- Model evaluation is broadly classified into two categories: internal evaluation and external evaluation.
- Internal evaluation assesses the model using the exact same dataset upon which it was trained.
- In external evaluation, the model is evaluated against a benchmark or gold standard, for example a set of objects with correct labels assigned by human experts; hence, this approach is also called supervised evaluation in contrast to the unsupervised approach used in internal evaluation.

Model Evaluation

- Standard internal evaluation criteria prioritize models that attain high intra-cluster similarity (i.e. high similarity between objects within a cluster) and low inter-cluster similarity.
- However, achieving high scores on internal evaluation does not inherently guarantee good performance on new data.
- Widely utilized internal evaluation measures for clustering include the Calinski-Harabasz index (variance ratio criterion), the Davies-Bouldin index, the silhouette coefficient, and the Gap statistic.
- Comparative evaluations generally highlight the Gap statistic and the Calinski-Harabasz index as the top performers among these metrics.

Model Evaluation

- For a dataset of N objects partitioned into K clusters, the Calinski–Harabasz index is defined by a variance ratio

$$I_{CH} = \frac{SS_B/(K-1)}{SS_W/(N-K)} \qquad (7)$$

where $SS_B/(K-1)$ is the overall between-cluster variance and $SS_W/(N-K)$ is the overall within-cluster variance. The between-cluster sum of squares, SS_B, is given by:

$$SS_B = \sum_{k=1}^{K} n_k \|c_k - c\|^2 \qquad (8)$$

where the kth cluster containing n_k data points has mean position c_k, and c is the mean of all sample data, with $\|\ldots\|$ denoting the Euclidean distance norm.

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Model Evaluation

- Similarly, the within-cluster sum of squares, SS_W, is given by:

$$SS_W = \sum_{k=1}^{K} \sum_{x \in C_k} \|x - c_k\|^2 \qquad (9)$$

where $\boldsymbol{x}$ is a data point in the kth cluster with mean position at c_k.

- Because the primary objective is to identify clusters exhibiting a large SS_B and small SS_W, one would search for the model with the largest I_{CH}, that is, when evaluating multiple models with varying K values, the optimal number of clusters is given by the model with the largest I_{CH}.
- For internal evaluation utilizing the Gap statistic, the sum of pairwise distances for all data points in a cluster (denoted by C_k) is given by:

$$D_k = \sum_{i,j \in C_k} d_{ij}, \qquad (10)$$

with the squared Euclidean distance commonly used for the distance measure d_{ij}.

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Model Evaluation

- The pooled within-cluster sum of squares around the cluster mean is given by,

$$W_K = \sum_{k=1}^{K} \frac{1}{2n_k} D_k \qquad (11)$$

(where the factor of 2 compensates for the double counting in D_k).

- Since W_K quantifies the compactness of the clustering, a small value of W_K is desirable.
- The gap statistic is simply the difference between $\log(W_K)$ and the expected value from a null reference distribution, that is, a distribution with no obvious clustering,

$$\text{Gap}(K) = E_N^*[\log(W_K)] - \log(W_K) \qquad (12)$$

where E_N^* represents the expectation under a sample of size N from the reference distribution.

Model Evaluation

- $E_N^*[\log(W_K)]$ is estimated via Monte Carlo sampling (B times from the null distribution, which can be the uniform distribution over the range of the observed data.
- From the B Monte Carlo samples, B values of $\log(W_K)$ were obtained, and were averaged to get the estimate of $E_N^*[\log(W_K)]$.
- Let sd_K denote the standard deviation computed from the B values of $\log(W_K)$. A standard error s_K is defined by:

$$s_K = sd_K\sqrt{1 + (1/B)} \qquad (13)$$

- Finally, the optimal K value is chosen to be the smallest K satisfying

$$\text{Gap}(K) \geq \text{Gap}(K+1) - s_{K+1} \qquad (14)$$

Model Evaluation

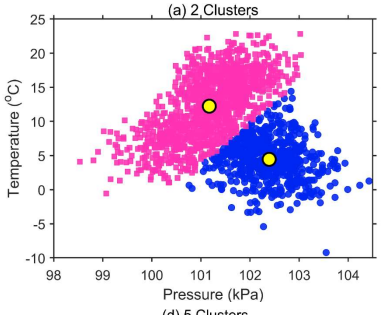
- Thus, we proceed from small to large K, stopping at the first K value that has $\mathrm{Gap}(K)$ exceeding $\mathrm{Gap}(K+1)$ within a tolerance of one standard error.
- The gap approach is computationally expensive because the clustering method needs to be applied to each of the B Monte Carlo samples.
- The gap statistic was refined so that the optimal K value is chosen to be the smallest K satisfying:

$$\mathrm{Gap}(K) \geq \mathrm{Gap}_{\max} - s_{\max} \qquad (15)$$

where $\mathrm{Gap}_{\max}$ is the global maximum of Gap found among all the K values tested, and $s_{\max}$ is the standard error at $\mathrm{Gap}_{\max}$.

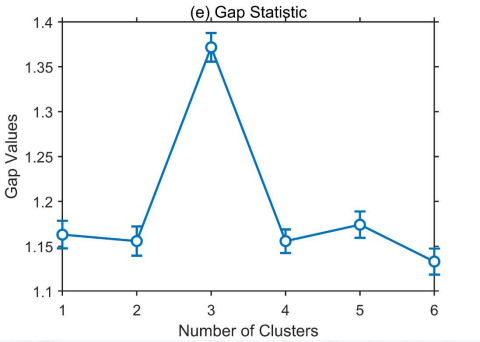
- The figure (in next slide) demonstrates K-means clustering with K = 2, 3, 4 and 5 clusters applied to a synthetic daily air pressure and temperature data at a location (5 years).

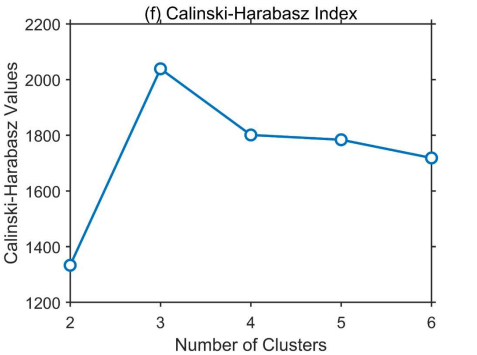
Model Evaluation



The pressure and temperature data have very different units, so it is necessary to perform clustering on the standardized data.

Model Evaluation





- To determine the optimal number of clusters, both the Gap statistic (15) and the Calinski-Harabasz index (7) were employed (the gap statistic can even be used for testing $K = 1$, whereas the Calinski–Harabasz index only works for $K \geq 2$).
- Ultimately, both evaluation statistics converged on the same result, identifying $K = 3$ as the optimal clustering configuration.

Model Evaluation

- External evaluation: The cluster model output is evaluated against a dataset with correct labels (benchmark).
  - Let $C = \{C_1, \ldots, C_K\}$ be the set of clusters from the model and $G = \{G_1, \ldots, G_{K'}\}$ be the clusters from the benchmark. For a pair of data points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$, there are four possible cases:
    - Case 1: $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ belong to the same cluster in C and to the same cluster in G.
    - Case 2: $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ belong to the same cluster in C but to different clusters in G.
    - Case 3: $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ belong to the different clusters in C but to the same cluster in G.
    - Case 4: $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ belong to the different clusters in C and to different clusters in G.

Model Evaluation

- Let a, b, c and d denote the number of pairs of points belonging to case 1, 2, 3 and 4, respectively. The total number of pairs of points is $M = a + b + c + d$, with M given by the binomial coefficient:

$$M = \binom{N}{2} = \frac{N(N-1)}{2} \quad (16)$$

- Some of the common indices used to evaluate the cluster model relative to the benchmark data.

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Model Evaluation

- The Rand index is given by,

$$R = \frac{a+d}{a+b+c+d} = \frac{a+d}{M} \quad (17)$$

where $a + d$ gives the number of pairs of data points where the cluster model is in agreement with the benchmark, and $0 \leq R \leq 1$.

- For large datasets with many clusters, a can be significantly overshadowed by d.
- When $d \gg a$, R becomes insensitive to a.

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Model Evaluation

- Because it is preferable for an evaluation metric to remain sensitive to a rather than being sensitive to d, it is often advantageous to delete d from the R index, resulting in the Jaccard index,

$$J = \frac{a}{a+b+c} \qquad (18)$$

- Another index that ignores d is the Fowlkes and Mallows index:

$$I_{FM} = \sqrt{\frac{a}{a+b}\frac{a}{a+c}} \qquad (19)$$

- Any of the aforementioned evaluation indices can be normalized with respect to a reference distribution.





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Statistical and Machine Learning Methods: Basics and Applications

MODULE 11: Support Vector Machines and Clustering

LECTURE W11L5: Clustering – Part II

BY

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CONCEPTS COVERED

Clustering

- ***K*-means Clustering**
- **Examples**

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K-means Clustering

- K-means clustering is an unsupervised learning approach categorized under non-hierarchical clustering.
- The objective of K-means clustering is to group the N data points ($\mathbf{x}_1$, $\mathbf{x}_2$,..., $\mathbf{x}_N \in \mathbb{R}^m$) into K sets or clusters ($C_1, C_2, \ldots, C_K$), which minimize the within-cluster sum of squares (i.e. squared Euclidean distance),

$$\sum_{k=1}^{K} \sum_{x \in C_k} |x - c_k|^2$$

where c_k is the centroid or centre of the cluster C_K, computed by taking the mean of all the data points belonging to the cluster.

- The standard algorithm to solve K-means clustering is Lloyd's algorithm.

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K-means Clustering

- The following procedure is employed to perform K-means clustering (Lloyd's algorithm):
  I. Initially, choose K, the number of clusters.
  II. Start with initial guesses for the mean positions of the K clusters in the x space (i.e. the position of a cluster centre, a.k.a. centroid, is simply the mean position of all the data points belonging to that cluster).
  III. Repeat the subsequent two steps until convergence:

  Step–I: For every data point, find the closest centroid (based on the squared Euclidean distance) and assign the data point to be a member of that cluster, where the Euclidean distance is given by: $d_{ij} = \sum_{l=1}^{m} \left(x_{il} - x_{jl}\right)^2$ with x_{il} and x_{jl} $(l = 1, \ldots, m)$ denoting the elements of $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

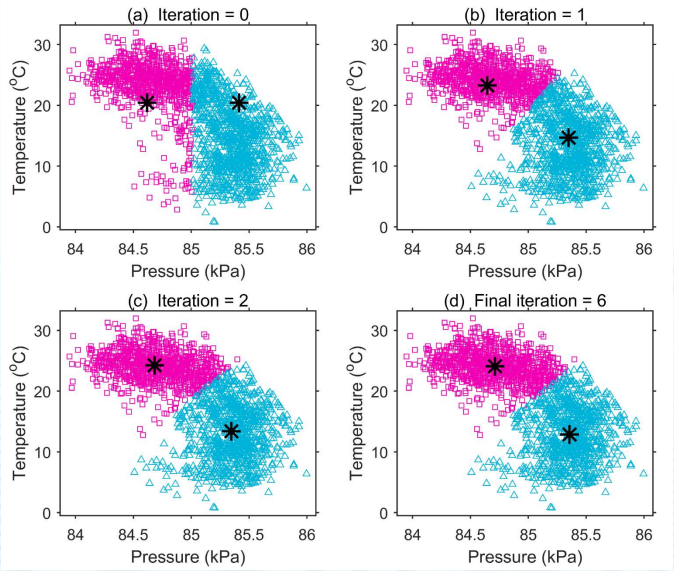
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K-means Clustering

Step – II: For each cluster, estimate the new centroid to be the mean position of all data points belonging to that cluster.

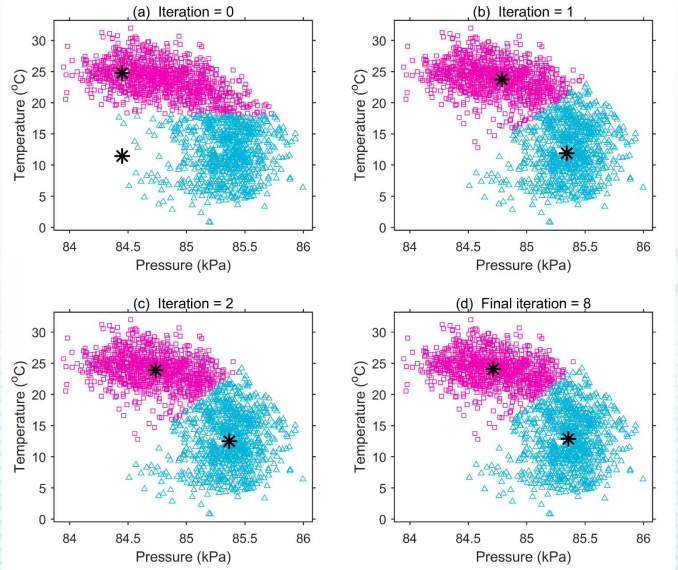
- The initial choice for the K centroids often involves randomly picking K data points from the x data.
- However, the effect of this initial random selection dies down with iterations and the finally identified clusters are irrespective of the initial choice.

K-means Clustering



The given figure illustrates K-means clustering with $K = 2$ clusters for the daily air pressure and temperature data at Dharamshala, India during 2013–2017.

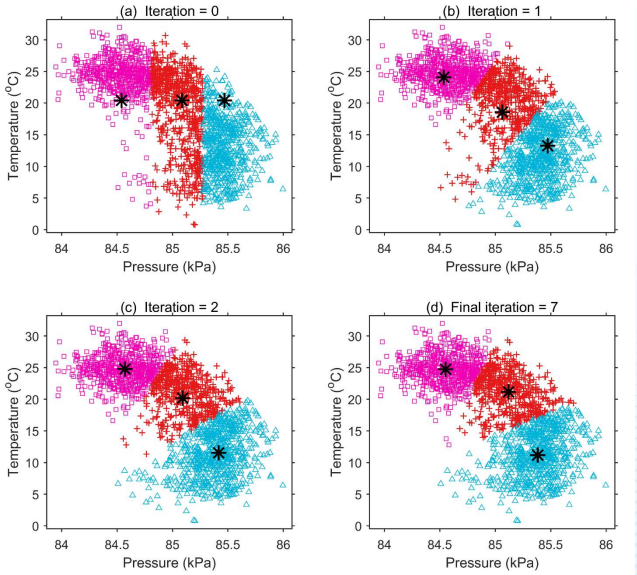
K-means Clustering



Same data (as shown in previous slide) but the initially selected centroids are different.

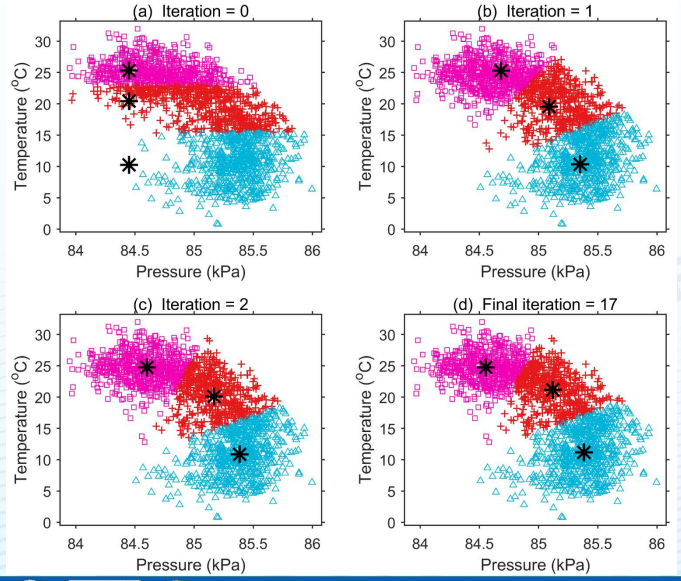
However, the final ones are same.

K-means Clustering



Same data (as shown in previous slide) but the initially selected centroids are more (k=3).

K-means Clustering



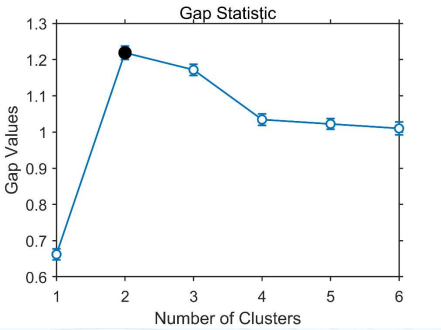
Same data (as shown in previous slide) but the initially selected centroids are different.

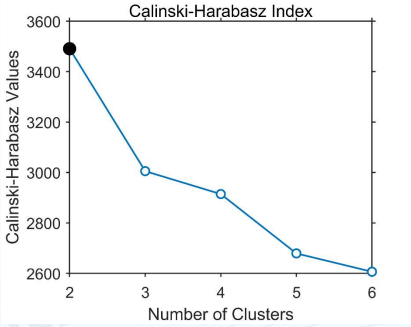
However, the final ones are same.

K-means Clustering

- Figure (a) displays the initial guesses for the centroids indicated by asterisks. The data points are allocated to clusters based on their nearest centroid.
- The figure (b) illustrates the recalculation of centroids derived from the mean position of the cluster members established in (a), and cluster members in (b) have been reassigned based on their proximity to the newly identified centroids in (b).
- The location of the centroids and their associated cluster members are shown after (c) two iterations and (d) after final convergence of the K- means clustering algorithm.
- The poor initial choice of the centroids did not hinder the convergence of the clustering algorithm.

K-means Clustering





- To determine the optimal number of clusters, both the Gap statistic and the Calinski-Harabasz index were employed.
- Ultimately, both evaluation statistics converged on the same result, identifying $K = 2$ as the optimal clustering configuration.

K-means Clustering

- In terms of computational expense, the operation count for one iteration of Lloyd's algorithm is $O(NKm)$.
- Since K and m are usually much smaller than N, this is considered $O(N)$, which is relatively fast among clustering methods.
- A primary drawback of K-means is that it often converges to a local minimum rather than the global minimum.
- Alternative strategies for avoiding poor local minima include using hill climbing or some type of evolutionary algorithm (e.g. genetic algorithm).
- The optimal K is determined by running the cluster model multiple times with different K values and using either internal or external evaluation to select the best model (as explained in the previous lecture).

Example – 1

Consider a dataset consisting of four points with standardized features x_1 and x_2 given in table.

Apply K-means clustering with $K = 2$ using squared Euclidean distance. Let the initial centroids be: $C_1 = (1,1)$, and $C_2 = (2,1)$.

Perform the iterations until the cluster assignments no longer change. Determine the cluster membership of points after each iteration and the final clusters.

Point	x_1	x_2
P_1	1	1
P_2	2	1
P_3	4	3
P_4	5	4

Solution:

We calculate the distance between each point and the centroids.

For computational efficiency in K-means, the squared Euclidean distance is used:

$$d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2$$

Iteration 1: Using $C_1 = (1,1)$ and $C_2 = (2,1)$

For $P_1(1,1)$:

Distance to C_1: $d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2 = (1-1)^2 + (1-1)^2 = 0$

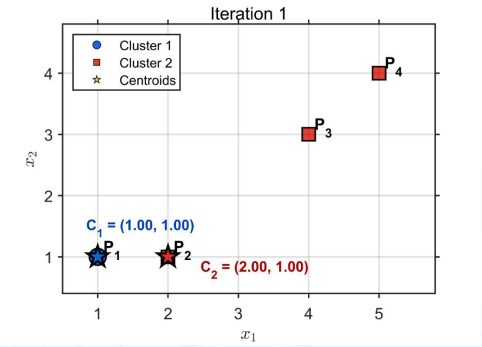
Distance to C_2: $d^2 = (x_1 - c_1)^2 + (x_2 - c_2)^2 = (1-2)^2 + (1-1)^2 = 1$

Since, Distance to C_1 < Distance to C_2, so, the assigned cluster is C_1.

Similarly, for all the other points calculations were carried out and results are tabulated.

Solution:

Point	$d^2(C_1)$	$d^2(C_2)$	Cluster Assigned
P_1	0	1	C_1
P_2	1	0	C_2
P_3	13	8	C_2
P_4	25	18	C_2



Iteration 2:

Now, new centroids are calculated as:

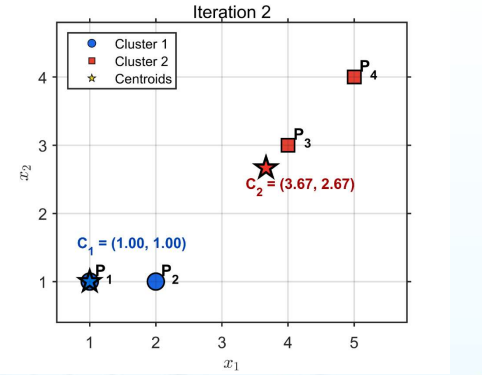
$C_1 = (1,1)$ and $C_2 = \left(\frac{2+4+5}{3}, \frac{1+3+4}{3}\right) = \left(\frac{11}{3}, \frac{8}{3}\right) = (3.67, 2.67)$

The calculations are again repeated for all the points considering the new centroids and results are tabulated.

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Solution:

Point	$d^2(C_1)$	$d^2(C_2)$	Cluster Assigned
P_1	0	9.89	C_1
P_2	1	5.56	C_1
P_3	13	0.22	C_2
P_4	25	3.56	C_2



Iteration 3:

Now, new centroids are calculated as:

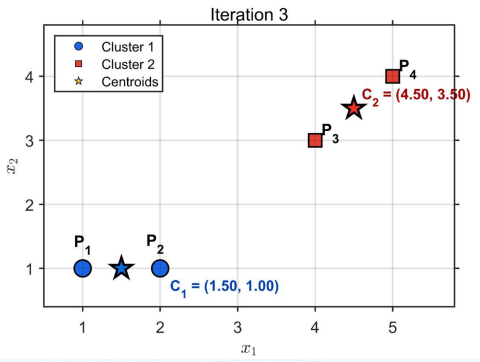
$C_1 = \left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1.5, 1)$ and $C_2 = \left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4.5, 3.5)$

The calculations are again repeated for all the points considering the new centroids and results are tabulated.

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Solution:

Point	$d^2(C_1)$	$d^2(C_2)$	Cluster Assigned
P_1	0.25	18.5	C_1
P_2	0.25	12.46	C_1
P_3	10.24	0.5	C_2
P_4	21.25	0.5	C_2



Because none of the pixel assignments changed, the algorithm has successfully converged. The final clusters are:

Cluster 1: $\{P_1, P_2\}$ and Cluster 2: $\{P_3, P_4\}$.





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ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 11: Support Vector Machines and Clustering

LECTURE W11L6: Clustering – Part III

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CONCEPTS COVERED

Clustering

- **Hierarchical Clustering**
- **Examples**

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Hierarchical Clustering

- Hierarchical clustering (HC) methods not only group objects (i.e. data points) into clusters but also arrange the clusters into a hierarchical structure, with the shape of an inverted tree.
- Two primary methodologies exist for hierarchical clustering: agglomerative HC (a bottom-up approach to building the inverted tree structure) and divisive HC (a top-down approach).
- In agglomerative HC, clusters are merged; that is, starting with small clusters each containing a single data point, two closest clusters are merged and the process is repeated until reaching a single cluster containing all the data points.
- The divisive approach initiates the process with a single, massive cluster containing the entire dataset, subsequently splitting it into progressively smaller constituent clusters.

Hierarchical Clustering

- Because the agglomerative method is significantly more computationally efficient than the divisive method, our focus will be exclusively on the former.
- There are several choices for the linkage measure, used in deciding which two clusters are close enough to justify merging them in agglomerative HC.
- The linkage measure determines the 'distance' between two clusters.
- All pairs of clusters have their distance determined by the linkage measure, and the pair with the smallest linkage value is merged.
- The most frequently implemented linkage measures include:
  - Maximum distance or complete linkage defines the distance between two clusters A and B as the maximum distance between a pair of data points, with one point belonging to A and the other to B,

Hierarchical Clustering

$$d(A,B) = \max_{x_i \in A, x_j \in B} d_{ij} \quad (1)$$

where d_{ij} is a distance measure for two data points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

- Minimum distance or single linkage defines the distance between two clusters A and B as being the minimum distance between a pair of data points, with one point in A and the other in B,

$$d(A,B) = \min_{x_i \in A, x_j \in B} d_{ij} \quad (2)$$

- Average linkage defines the distance between clusters A and B as the average distance between all possible pairs of points, with one point in A and the other in B,

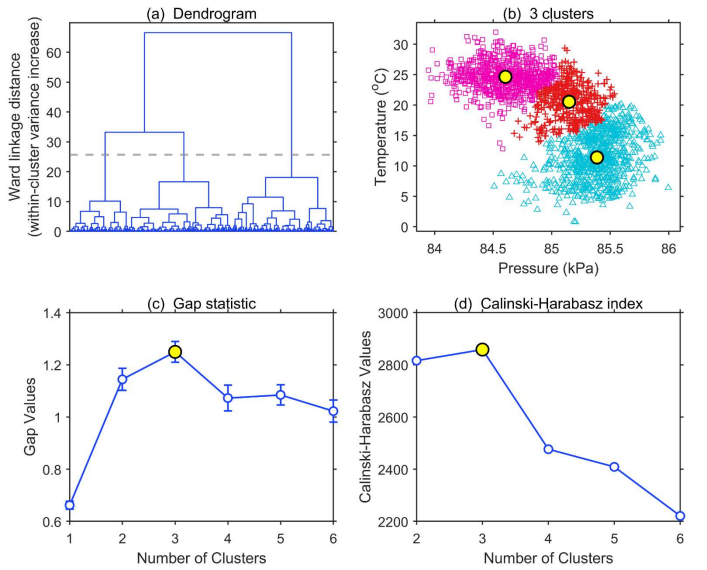
$$d(A,B) = \frac{1}{n_A n_B} \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} d_{ij} \quad (3)$$

with n_A and n_B being the number of points in cluster A and B.

Hierarchical Clustering

- Another popular choice is Ward's method.
- At each step, this approach finds the pair of clusters that, when merged, would lead to the minimum increase in the total within-cluster sum of squared Euclidean distance SSW (as defined in previous lecture).
- The resulting inverted-tree hierarchy can be visually represented using a dendrogram.
- A dendrogram consists of many '⊓'-shaped lines connecting two clusters in a hierarchical tree (figure in next slide).
- The vertical axis displays the distance measure of two clusters (or in the case of Ward's methods, the SSW).

Hierarchical Clustering



The figure represents – Hierarchical clustering of the daily air pressure and temperature data at Dharamshala, India using Ward's method on standardized data. (a) Dendrogram, where the cut-off level (horizontal dashed line) is set to three clusters, (b) the three clusters, (c) gap statistic and (d) Calinski–Harabasz index.

Hierarchical Clustering

- The length of the shorter vertical line in ⊓ represents the distance between the two clusters to be merged.
- Adjusting the cut-off level represented by the horizontal dashed line in the dendrogram, one can vary the number of clusters retained.
- Here the cut-off is set at three clusters, as shown in the figure (previous slide).
- The optimal number of clusters is $K = 3$ as chosen by the Calinski–Harabasz index and Gap statistic.
- Compared with the K-means clustering with $K = 3$, the three clusters obtained using Ward's method exhibit more jagged boundaries.
- The maximum Calinski–Harabasz index at $K = 3$ is nearly 2860 for Ward's method and nearly 3500 for K-means at $K = 2$.

Hierarchical Clustering

- The Calinski-Harabasz index is the ratio of the overall between-cluster variance to the overall within-cluster variance; thus, a lower ratio for Ward's method indicates a poorer clustering result compared with K- means.
- Furthermore, the gap statistic selects $K = 3$ because the standard error bar is wider for Ward's method (as shown in panel-c) than for K-means, which reflects the poor robustness of HC methods.
- The common criticism for classical HC algorithms is that they lack robustness and are, hence, sensitive to noise and outliers.
- Once an object is assigned to a cluster, it will not be considered again, which means that HC algorithms are not capable of correcting possible previous misclassification.
- With sample size N, HC algorithms are $O(N^2)$ versus $O(N)$ for K-means, hence, HC is slower than K-means clustering for large datasets.

Example – 2

Four observations having two features in each are given in the following table. Using Euclidean distance and single-linkage agglomerative hierarchical clustering,

i) show the order in which the observations/clusters are merged and state the linkage distance at each merging step.

ii) determine the resulting clusters if the dendrogram is cut at a distance of 3.5.

Observation	x_1	x_2
A	1	1
B	1	3
C	5	4
D	8	4

Solution:

The Euclidean distance between two points is given by:

$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2}$$

Step 1: Pair-wise Distances

For A (1,1) and B (1,3):

$$d_{AB} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(1-1)^2 + (1-3)^2} = 2$$

Similarly, For A (1,1) and C (5,4):

$$d_{AC} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(1-5)^2 + (1-4)^2} = 5$$

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Solution:

Similarly, For A (1,1) and D (8,4):

$$d_{AD} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(1-8)^2 + (1-4)^2} = 7.62$$

For B (1,3) and C (5,4):

$$d_{BC} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(1-5)^2 + (3-4)^2} = 4.12$$

Similarly, For B (1,3) and D (8,4):

$$d_{BD} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(1-8)^2 + (3-4)^2} = 7.07$$

Similarly, For C (5,4) and D (8,4):

$$d_{AC} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2} = \sqrt{(5-8)^2 + (4-4)^2} = 3$$

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Solution:

Step 2: First Merge

Initially, all the points are considered as separate clusters $[\{A\},\{B\},\{C\},\{D\}]$.

The smallest distance is $d_{AB} = 2$.

Therefore, A and B merge first to form the cluster $\{A, B\}$. The linkage distance is 2.

Step 3: Finding the next closest clusters

Till now, we have the clusters – $[\{A, B\},\{C\},\{D\}]$.

For single linkage,

$$d(\{A, B\}, C) = \min[d(A, C), d(B, C)] = \min(5, 4.12) = 4.12$$

And,

$$d(\{A, B\}, D) = \min[d(A, D), d(B, D)] = \min(7.62, 7.07) = 7.07$$

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Solution:

Also, $d(C, D) = 3$.

The smallest distance is therefore, $d_{CD} = 3$.

Hence, C and D merge first to form the cluster $\{C, D\}$. The 2nd linkage distance is 3.

Step 4: Final Merge

Now, only two clusters remain – $[\{A, B\},\{C, D\}]$.

Using single linkage,

$$d(\{A, B\},\{C, D\}) = \min\begin{cases} d(A, C), \\ d(A, D), \\ d(B, C), \\ d(B, D) \end{cases} = \min(5, 7.62, 4.12, 7.07) = 4.12$$

So, the final merge is $\{A, B\}$ and $\{C, D\}$ at a linkage distance of $d = 4.12$.

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Solution:

Cut-Off at $d = 3.5$

At $d = 3.5$:

- A and B have already merged at $d = 2$.
- C and D have already merged at $d = 3$.
- But, the 2 clusters do not merge until $d = 4.12$.

Therefore, the dendrogram contains two clusters at the cut-off:

$$C_1 = \{A, B\} \quad \text{and} \quad C_2 = \{C, D\}$$

